

is why a rollout plan having flexible hardware with backward compatibility to legacy technologies is a prime requirement for the implementation of new wireless standards. "This calls for devices and infrastructure equipment supporting multi-technology with flexible plugins of hardware and software modules," says Jayanth Ramachandran, business development head - telecom, Rohde & Schwarz India.

The software and hardware should be smart, proactive and reduce CPU usage of a device. Custom-developed application-specific integrated circuits (ASICs) are being used to solve these challenges. The platform also needs to support both frequency division duplexing (FDD) and time division duplexing (TDD) modes, depending on the spectrum used. "In the case of software, upgrades should be possible without stopping the device," comments Nagaraja.

The hardware approved for a foreign country may not work in India due to the environmental conditions here. Any generic technology is to be customised for matching the Indian scenario, and there should be a well-defined set of standards for testing and manufacturing of the same. Field testing of telecom equipment, which includes interoperability development testing (IODT) and interoperability testing (IOT), are necessary before bringing any product to the market. But, due to prevailing uncertainties related to frequency bands, test and measurement firms will have to support a wide range of bands and antennae schemes.

"However, one of the biggest concerns for implementing 5G technology would be the uncertainty of the propagation model at higher frequency bands, which are being planned to be used," comments Jayanth. Different regions of the spectrum are suited for different applications. Ravi Manik, director of business development at Broadcom adds, "The end user experience depends on what kind of bandwidth is available."

Two distinct themes for 5G

1. A super-fast mobile network comprising next-generation small cells, densely clustered together to give a contiguous coverage over at least urban areas and delivering peak data rates of up to 1Gbps. It gets the world to the final frontier for true wide area mobility. Spectrum for this would be under 4GHz (in order to make wide area coverage economically feasible). Since nobody is likely to reserve spectrum for 5G, such a network would enter the market through advanced spectrum sharing. Hence, in this vision 5G might be the first global implementation of dynamic spectrum access. The band 3.4GHz - 3.6GHz has been talked about for such a spectrum sharing market entry for 5G.

2. A converged wireless fibre network that uses, for the first time for wireless Internet access, millimetre wave bands (20GHz - 60 GHz) so as to allow very wide bandwidth radio channels to be able to support data access speeds of up to 10Gbps. The connection essentially comprises short wireless links at the end of the local fibre optic cable. It takes data over wireless to state-of-the-art level. But this very high data speed comes at a price for consumers in terms of very much reduced coverage. It would be more of a nomadic service (like Wi-Fi) to fixed locations rather than a wide area mobile service.

—Courtesy: GSMHistory.com

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Defence sector occupies a key portion of spectrum in India, and firms feel that they need to be funded by the government to vacate and release additional (unused) spectrum for the development of the commercial telecom sector in our country.

Alternate bands. Commercialisation of television white spaces can decongest the existing 3G and existing 4G networks. With superior wireless radio propagation characteristics, these can also lead to better mobile coverage in rural areas. Designers are also looking into millimetre wave and microwave technologies to solve the challenges of backhaul connectivity in mobile networks, according to Manik.

What are white spaces

White spaces refer to the unlicensed frequencies in the spectrum allocated to broadcasting services. With television broadcast moving from analogue to digital, a lot of frequency bands get freed up; these unused frequen-

cies are known as television white spaces. The average white space in the ultra high frequency (UHF) television band in India is above 100MHz (as per quantitative assessment of television white space in India by Gaurang Naik, Sudesh Singhal, Animesh Kumar and Abhay Karandikar, IIT Bombay.) This technology is still in its innovation stage and several prototypes for power over Ethernet (PoE) system using this method are being tried out.

Better-connected future

With the ever-rising number of smart-devices and an increasing demand for better quality of service, telecommunication and related technologies are being researched and remodelled continuously. While work is being carried out to improve existing systems and make these accessible to the general public, there is no doubt that 5G is going to be the next big thing in communication and can be expected to reach the market in another five years. ●